

Fundamental Principles of Quantum Theory.

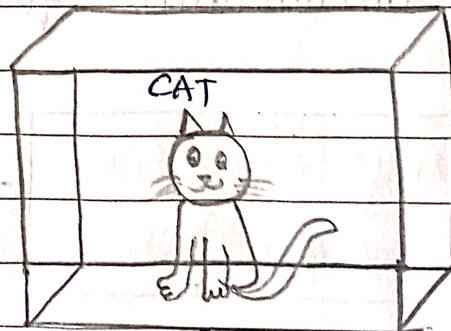
→ SUPERPOSITION: Schrodinger at 1926 gave a powerful equation that explains how quantum particles behave like waves, leading to the concept of Superposition.

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

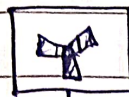
→ According to the principle of superposition, a quantum particle can exist in multiple possible states simultaneously until it is measured.

→ In view of classical thinking the switch is either ON(1) or OFF(0). In terms of quantum thinking a qubit can be 0, 1, both 0 and 1 at the same time.

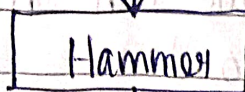
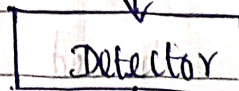
Experimentation: - Schrodinger's cat. is a thought experiment used to explain quantum superposition and the measurement problem in quantum mechanics.



Box.



Radio active Atom.



poison bottle

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→ setup of the Experiment, a sealed box contains
(i) a cat, a radioactive atom, a geiger counter, a hammer,
a poison bottle.

① → If the atom decays, the geiger counter detect it and
the hammer breaks poison bottle and the cat dies.

② → If the atom does not decay, the poison is not released
and cat remains alive.

Quantum
concept

Before opening the box, the atom is in a superposition
of decayed and undecayed states. Therefore, the cat
is considered both alive and dead simultaneously. The purpose
of this experimentation is to criticize Copenhagen Interpretation.
This shows the difficulties of applying quantum rules in everyday
objects.

Real Time
Example

→ Spinning Coin

→ Before stopping it appears both heads and tails

→ After observation it becomes either heads or tails.

Therefore

→ Schrodinger's cat is a thought experiment proposed by
Erwin Schrodinger in 1935. According to the quantum
mechanics, before observation the cat exists in a superposition
of alive and dead states. This experiment illustrates quantum
superposition and the measurement problem.

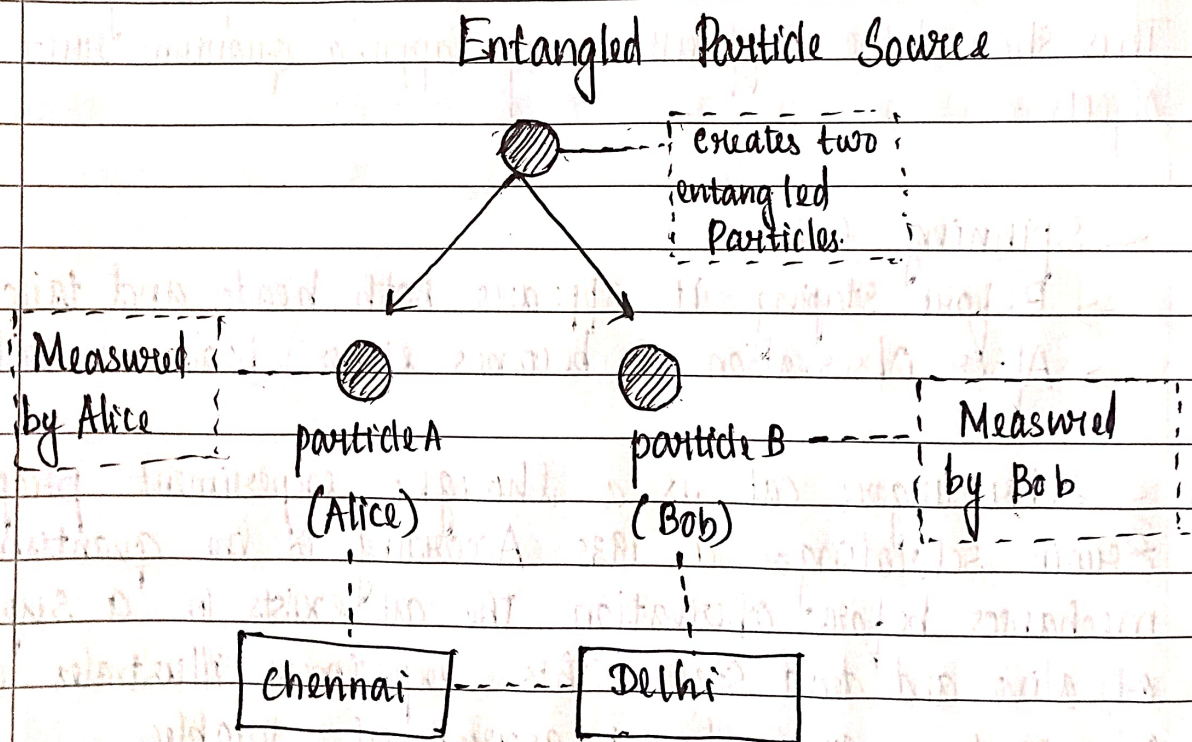
2. → ENTANGLEMENT (1935): Albert Einstein, Boris Podolsky, Nathan Rosen

→ Quantum entanglement is a phenomenon in which two or more particles become connected so that the state of one particle is related to the state of the other, even when they are separated by large distances.

→ EPR Paradox:

→ Einstein, Podolsky, and Rosen proposed the EPR (Einstein-Podolsky-Rosen) Paradox to challenge quantum mechanics.

→ They questioned: Can two distant particles remain connected and influence each other instantly. Einstein called this phenomenon called Spooky Action at a Distance.



1. → An entangled particle source creates two entangled particles

2. → The particles move in opposite directions

- Particle A → Alice (Chennai)

- Particle B → Bob (Delhi)

3 → Alice measures Particle A, suppose she gets

Particle A = $\uparrow$ (spin up)
Immediately she knows,

Particle B = $\downarrow$ (spin down), even though Particle B is far away.

→ Why was it a Paradox? According to classical physics

→ objects have definite properties

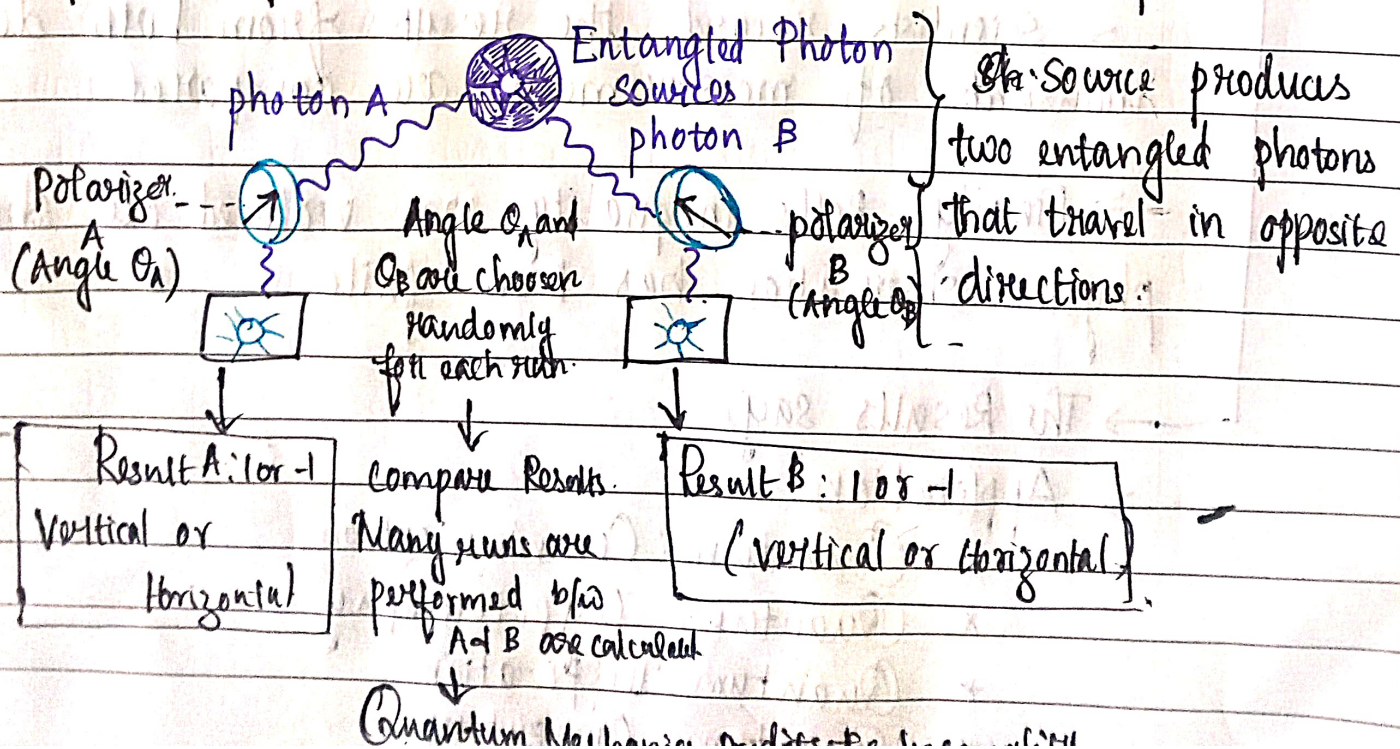
→ Information cannot travel faster than light

→ But entanglement suggested that the particles remain correlated regardless of distance. This puzzled Einstein.

→ BELL TEST EXPERIMENT (1964) by John Bell

→ Bell Proposed this test, because the EPR Paradox raised an important question: $\Rightarrow$ Are the particles carrying hidden information before measurement, or is quantum entanglement real?

→ To answer this, John Bell developed Bell's theorem and proposed experiments known as Bell test Experiments.



Reference.

* Detector is an instrument used to observe, measure, and record physical phenomena - most commonly subatomic particles, radiation or energy. By capturing the interactions of these particles, detectors help physicists determine properties like mass, charge, momentum, and energy.

* A 'polarizer' is an optical filter that lets light waves of a specific direction pass through while blocking light waves vibrating in other orientations.

→ The Bell experimentation results, that

- * Quantum mechanics predictions were correct
- * Entanglement exists.
- * classical explanations were sufficient.

→ From the experimental diagram, a source produces two entangled photons.

Source → Photon A + Photon B

→ The photons travel in opposite directions. Each photon passes through a measuring device called a polarizer.

→ Scientists record the results from both detectors. Thousands of measurements are compared.

→ Finally, Bell's inequality was violated (i.e. classical physics observation has been false).

→ The Results say,

→ Application

- * Quantum Communication
- * Quantum Cryptography
- * Quantum Teleportation
- * Quantum Computation